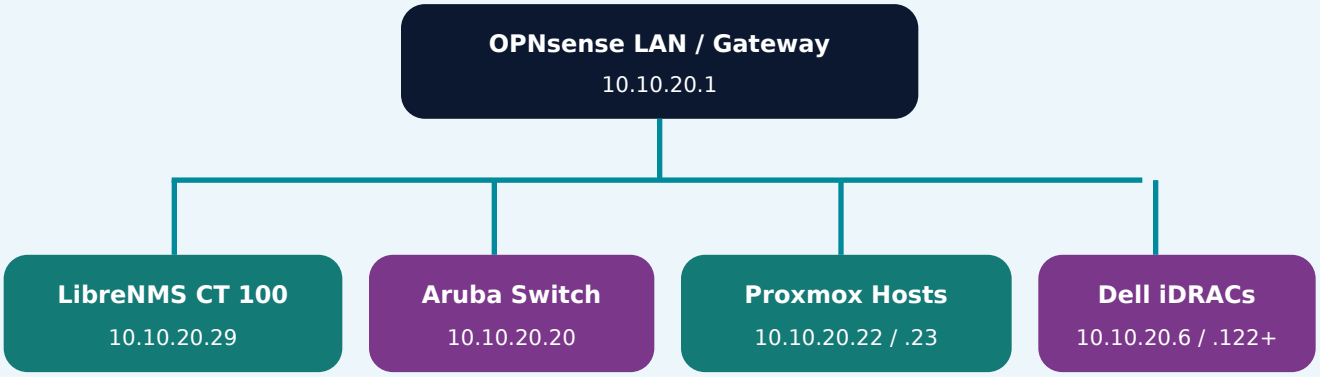
NETWORK SUPPORT SERVICES • MOCKTIA FIELD GUIDE

LibreNMS in Proxmox

Complete setup, SNMP discovery, OPNsense firewall, Dell iDRAC and troubleshooting

A companion to the interactive MockTIA lab. Designed for real hands-on setup in a Debian 13 LXC running on Proxmox and a controlled 10.10.20.0/24 network.



LibreNMS polls devices over UDP 161 (SNMP) and checks ICMP reachability.

TCP 80/443 is for the LibreNMS web dashboard. This topology is a training-lab reference.

Lab parameter	Value
Container	CT 100, Debian 13, host name LibreNMS
Network	10.10.20.29/24; gateway 10.10.20.1
Resources	2 vCPU, 4 GB RAM recommended, 32 GB disk
Monitored devices	Proxmox .22/.23; Aruba .20; OPNsense .1; iDRAC .6/.122/.123/.125
Access	http://10.10.20.29 (lab only until HTTPS)

Student: _____ Date: _____

Created by Mr. Beaird • MockTIA - Interactive IT Training & Simulations

1. Objectives, safety and accounts

By the end of the practical, the student can deploy and validate LibreNMS, distinguish monitoring protocols, configure SNMP on three types of network devices, scan the approved subnet and diagnose common failures.

Accounts: same names, different permission boundaries

Account	Used where	Purpose
root	Proxmox shell / CT console	System packages, service management, configuration files
librenms (Linux)	Debian 13 CT shell	Runs poller, composer, scans and validate.php
dbeaird (web)	LibreNMS browser	Dashboard user: devices, credentials, alerts, graphs
librenms (MariaDB)	Database connection	Database-only account; not a Linux or browser login

TCP/IP and port reference

Traffic	Port / Protocol	Reason
Browser	80/TCP; later 443/TCP	LibreNMS interface
SNMP polling	161/UDP	LibreNMS queries agents
SNMP traps	162/UDP (optional)	Devices notify trap receiver
Reachability	ICMP	Ping/fping, not the same as SNMP
DNS / downloads	53/UDP,TCP; 443/TCP	Resolve/install dependencies

Security boundaries

- Use public only for the isolated student exercise. It is widely known and transmitted without encryption in SNMPv2c.
- Restrict UDP 161 to source 10.10.20.29/32; use SNMPv3 for managed production networks.
- Do not expose the HTTP dashboard to the WAN; configure HTTPS before remote access.
- Back up the LibreNMS .env with its APP_KEY securely; never include it in student submissions.
- Scan only the authorized 10.10.20.0/24 lab subnet.

Deployment reminder

- Proxmox GUI → Create CT → Debian 13 → CT 100 → 2 vCPU / 4 GB RAM / 32 GB disk.
- Choose a bridge that reaches the lab; static address 10.10.20.29/24, gateway 10.10.20.1, lab DNS 10.10.20.1.
- Start the container. Root in the container is different from root in the Proxmox host.

2. Debian 13 installation procedure

Run these commands inside CT 100 as root unless otherwise marked. If the container is already installed, do not repeat the database/user creation steps.

Check networking and install dependencies

```
ip -4 -br addr
ip route
getent hosts deb.debian.org
apt update
apt install -y acl ca-certificates curl fping git lsb-release \
 mariadb-client mariadb-server mtr-tiny nginx-full nmap \
 php-cli php-curl php-fpm php-gd php-gmp php-mbstring \
 php-mysql php-snmp php-xml php-zip python3-command-runner \
 python3-dotenv python3-pip python3-psutil python3-pymysql \
 python3-redis python3-setuptools python3-systemd rrdtool \
 snmp snmpd unzip wget whois cron composer
```

Create the app account, clone and set permissions

```
useradd librenms -d /opt/librenms -M -r -s "$(which bash)"
cd /opt && git clone https://github.com/librenms/librenms.git
chown -R librenms:librenms /opt/librenms
chmod 771 /opt/librenms
setfacl -d -m g::rwx /opt/librenms/rrd \
 /opt/librenms/logs /opt/librenms/bootstrap/cache/ \
 /opt/librenms/storage/
setfacl -R -m g::rwx /opt/librenms/rrd \
 /opt/librenms/logs /opt/librenms/bootstrap/cache/ \
 /opt/librenms/storage/
```

Composer: use the librenms account

```
su - librenms
cd /opt/librenms
./scripts/composer_wrapper.php install --no-dev
# In blocked networks: install Debian composer as root first,
# then run the following as librenms:
composer install --no-dev --prefer-dist --no-interaction
ls -l /opt/librenms/vendor/autoload.php
exit
```

The exact missing-file error vendor/autoload.php caused HTTP 500 during this classroom setup. Never bypass TLS verification just to fetch Composer.

3. MariaDB, time zone, PHP and Nginx

MariaDB config and credentials (root inside CT)

```
cat > /etc/mysql/mariadb.conf.d/99-librenms.cnf <<'EOF'
[mariadbd]
innodb_file_per_table=1
lower_case_table_names=0
EOF
systemctl enable --now mariadb
systemctl restart mariadb
mariadb -u root
```

Inside the MariaDB prompt, substitute a strong, private database password:

```
CREATE DATABASE librenms
 CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;
CREATE USER 'librenms'@'localhost'
 IDENTIFIED BY 'CHANGE_ME_SECURELY';
GRANT ALL PRIVILEGES ON librenms.* TO 'librenms'@'localhost';
EXIT;
```

Match PHP, system and database time

```
timedatectl set-timezone America/Chicago
# Set date.timezone = America/Chicago in BOTH:
# /etc/php/8.4/fpm/php.ini
# /etc/php/8.4/cli/php.ini
cp /etc/php/8.4/fpm/pool.d/www.conf \
 /etc/php/8.4/fpm/pool.d/librenms.conf
# Edit librenms.conf:
# [www] -> [librenms]
# user = librenms ; group = librenms
# listen = /run/php-fpm-librenms.sock
systemctl restart php8.4-fpm
```

Nginx web root / PHP-FPM path

```
# Create /etc/nginx/sites-available/librenms.vhost
server {
 listen 80;
 server_name 10.10.20.29;
 root /opt/librenms/html;
 index index.php;
 location / { try_files $uri $uri/ /index.php?$query_string; }
 location ~ [^/]\.php(/|$) {
  fastcgi_pass unix:/run/php-fpm-librenms.sock;
  fastcgi_split_path_info ^(.+\.php)(/.+)$;
  include fastcgi.conf;
 }
 location ~ /\.(?!well-known).* { deny all; }
}
```

The Nginx server block above is a file configuration, not shell commands. After saving, enable it:

```
ln -sf /etc/nginx/sites-available/librenms.vhost \
 /etc/nginx/sites-enabled/librenms.vhost
nginx -t
# If syntax test succeeds:
rm -f /etc/nginx/sites-enabled/default
systemctl reload nginx
```

4. Browser installer and polling

Complete the LibreNMS installer

- Browse http://10.10.20.29/install.
- Confirm pre-install and database checks. MariaDB host is localhost, name/user are librenms, password is the DB password.
- Create the web administrator (example: dbeaird), with its own strong password.
- If the installer requests config.php, save its supplied contents and run: chown librenms:librenms /opt/librenms/config.php.

Configure cron, scheduler and log rotation (root)

```
cp /opt/librenms/dist/librenms.cron /etc/cron.d/librenms
chmod 644 /etc/cron.d/librenms
systemctl enable --now cron
cp /opt/librenms/dist/librenms-scheduler.service \
 /opt/librenms/dist/librenms-scheduler.timer \
 /etc/systemd/system/
systemctl daemon-reload
systemctl enable --now librenms-scheduler.timer
cp /opt/librenms/misc/librenms.logrotate \
 /etc/logrotate.d/librenms
systemctl list-timers --all | grep librenms
```

Validate the deployment

```
su - librenms -c 'cd /opt/librenms && ./validate.php'
systemctl is-active nginx mariadb php8.4-fpm cron
journalctl -u librenms-scheduler.service -n 30 --no-pager
```

Validation message	Interpretation
[OK] Database Connected	Application can reach MariaDB
[OK] Python poller wrapper is polling	Scheduled polling executed
[OK] MySQL and PHP time match	PHP time zone is aligned
[WARN] You have no devices	Normal before adding SNMP targets
[FAIL] Scheduler is not running	Inspect timer/service and journal

Proxmox LXC containers share the host clock. The host can synchronize NTP while the container reports its own NTP service as inactive.

5. SNMP: Proxmox hosts and OPNsense

Proxmox SNMP agent on .22 and .23

Run on each Proxmox HOST shell, not in the LibreNMS CT. Confirm the actual host IP first.

```
hostname
hostname -I
apt install -y snmpd snmp
cp -n /etc/snmp/snmpd.conf /etc/snmp/snmpd.conf.lab-backup
# Fresh lab agent configuration in /etc/snmp/snmpd.conf:
agentAddress udp:161
rocommunity public 10.10.20.29
sysLocation Proxmox Lab
sysContact Network Support Services
# As host root, after editing file:
systemctl enable --now snmpd
systemctl restart snmpd
ss -lunp | grep ':161'
```

The four SNMP config lines go inside /etc/snmp/snmpd.conf, not directly into the shell. For ongoing installations, merge with existing agent settings instead of overwriting custom monitoring configuration.

Test from LibreNMS CT (not from an unauthorized host source)

```
snmpget -v2c -c public 10.10.20.23 1.3.6.1.2.1.1.1.0
snmpget -v2c -c public 10.10.20.22 1.3.6.1.2.1.1.5.0
fping -4 -c 3 10.10.20.23
```

OID 1.3.6.1.2.1.1.1.0 is sysDescr.0 (system description); 1.3.6.1.2.1.1.5.0 is sysName.0 (SNMP hostname). A trailing .0 identifies a scalar instance.

OPNsense 10.10.20.1 (web GUI)

- System → Firmware → Plugins: install os-net-snmp when missing.
- Services → Net-SNMP → General: Enable; Community public; Listen IP 10.10.20.1; Apply.
- Firewall → Rules on the interface receiving traffic from 10.10.20.29: Pass UDP, source 10.10.20.29/32, destination This Firewall, port 161.
- Place rule ahead of matching blocks; optionally log matches and inspect Firewall → Log Files → Live View.

```
# On LibreNMS CT
fping -4 -c 3 10.10.20.1
snmpget -v2c -c public 10.10.20.1 1.3.6.1.2.1.1.1.0
```

6. Dell iDRAC hardware monitoring

iDRAC is the Dell Remote Access Controller. LibreNMS can poll hardware status even if the server OS is unavailable, provided the management NIC and iDRAC stay powered and reachable.

Configure the Dell iDRAC web GUI

- Open the current iDRAC management IP, e.g. https://10.10.20.6.
- For iDRAC7/8 look under iDRAC Settings → Network → Services; for iDRAC9 look under iDRAC Settings → Services. Menus vary by firmware.
- Enable SNMP Agent (not merely traps), choose All/v1-v2c where necessary, community public, discovery UDP port 161; save/apply.
- Verify static address 10.10.20.6, subnet mask 255.255.255.0, gateway 10.10.20.1, and the correct management VLAN.
- If IP changed from .122/.123/.125 to .6, identify which existing device entry corresponds to the Service Tag and update its LibreNMS device address.

Test SNMP and identify the hardware

```
# In CT 100:
snmpget -v2c -c public 10.10.20.6 1.3.6.1.2.1.1.1.0
snmpget -v2c -c public 10.10.20.6 1.3.6.1.2.1.1.5.0
# On a physical Dell Proxmox host:
dmidecode -s system-manufacturer
dmidecode -s system-product-name
dmidecode -s system-serial-number
dmidecode -s bios-version
```

Item	Purpose
sysName e.g. RAC_1JF01L1	iDRAC reports RAC name; suffix may match Dell Service Tag
dmidecode system-product-name	Identifies R620/R730/R740 etc.
dmidecode system-serial-number	Compare Service Tag to RAC suffix
iDRAC SNMP sensors	Fans, PSU, temperatures, RAID where available

Common fault: If fping 10.10.20.6 works but snmpget times out, check the iDRAC SNMP Agent and protocol/port/community. A corrected subnet mask alone does not guarantee SNMP is active.

7. Scan, verify and organize network devices

LibreNMS scans IP addresses using available credentials. It does not automatically enable SNMP on targets. The optional ping fallback can add many ICMP-only devices.

Configure allowed network; scan as librenms Linux user

```
su - librenms
cd /opt/librenms
./lnms config:get nets
# Add the authorized lab range if it is not already present:
./lnms config:set nets.+ '10.10.20.0/24'
./lnms scan:network 10.10.20.0/24
# OPTIONAL: add ping-only hosts that do not answer SNMP:
# ./lnms scan:network 10.10.20.0/24 --ping-fallback
```

Expected inventory (verify physical identity before renaming)

IP	Expected function
10.10.20.1	OPNsense firewall
10.10.20.20	Aruba 2930M switch
10.10.20.22	Proxmox installation-02
10.10.20.23	Proxmox installation-03
10.10.20.6	Dell iDRAC (recently changed management address)
10.10.20.122 / .123 / .125	Previously discovered Dell iDRAC addresses

Browser workflow

- Open http://10.10.20.29/addhost → enter IP → SNMP v2c → saved credential public → UDP 161.
- Go to Devices → All Devices and review discovery status.
- Choose descriptive display names (Proxmox2, Proxmox3, OPNsense-Firewall, Dell-R730-iDRAC) without changing a host OS name.
- Capture interface, hardware-health and uptime graphs after at least one successful poll cycle.

How to interpret scan results

A “no SNMP reply” result usually means no SNMP agent, incorrect credentials, wrong bind IP, port filtering, or device not reachable. A “no ping” result may mean no active device or blocked ICMP; it does not prove a physical device is absent.

8. Troubleshooting decision guide

Symptom	What to test and fix
Nginx welcome page	Inspect server_name and root /opt/librenms/html; nginx -t; disable default site
HTTP 500, vendor/autoload.php	Run Composer as Linux librenms; confirm file exists; inspect Nginx error.log
Composer signature failed	Check DNS, proxy, HTTPS CA; install Debian composer through apt and run dependencies
MariaDB credential failure	Test mariadb -u librenms -p librenms; database password differs from web login
PHP/MySQL six-hour mismatch	Compare date, PHP CLI and MariaDB NOW(); set PHP CLI/FPM America/Chicago
Python polling FAIL	Check /etc/cron.d/librenms; systemctl is-active cron; journalctl -u cron
Scheduler FAIL	Check systemctl status librenms-scheduler.timer and service journal
ping: Operation not permitted	Use fping as Linux librenms user; standard ping may lack CAP_NET_RAW
Proxmox3: xmt/rcv 0/0	Use IP 10.10.20.23; verify getent hosts Proxmox3; rerun fping with count
SNMP times out, ping OK	Confirm SNMP Agent, source ACL, community, UDP 161 and firewall log
Changed iDRAC subnet / IP	Check /24 mask, gateway, management VLAN, current IP, SNMP Agent; update existing record
Missing Proxmox cluster config	pvecm nodes failure indicates standalone host; clustering not needed to monitor

Single-pass diagnostic sequence inside LibreNMS CT

```
ip route get 10.10.20.23
fping -4 -c 3 10.10.20.23
snmpget -v2c -c public 10.10.20.23 1.3.6.1.2.1.1.0
getent hosts Proxmox3
su - librenms -c 'cd /opt/librenms && ./validate.php'
tail -n 40 /var/log/nginx/error.log
```

9. Vocabulary and sign-off

Term	Student definition
LXC container	OS-level isolated Linux guest managed by Proxmox
CT ID	Number identifying a Proxmox container (e.g. 100)
SNMP / MIB / OID	Monitoring protocol; object schema; unique object identifier
sysDescr.0 / sysName.0	System description and reported SNMP device name
SNMP v2c community	Plaintext shared read credential; not secure for production
SNMPv3	More secure SNMP authentication and encryption when configured
ICMP / fping	Network reachability test; not SNMP
MariaDB	Relational database storing LibreNMS configuration and measurements
PHP-FPM / Nginx	Application runtime / web server
Composer vendor/autoload.php	Dependency autoloader required for PHP app
cron / scheduler	Regular polling and application scheduled tasks
iDRAC / RAC	Dell out-of-band remote hardware controller
Service Tag	Dell identifier to match physical host to iDRAC
/24 subnet mask	255.255.255.0; local IPv4 subnet range

Student evidence and instructor validation

- □ CT 100 configuration (Debian 13, storage, IP and gateway)

- ☐ Pre-install / database checks and validate.php without FAIL entries
- ☐ SNMP test for Proxmox host, OPNsense firewall and one Dell iDRAC
- ☐ Network scan 10.10.20.0/24 and discovered devices screenshot
- ☐ One interface or hardware-health graph with explanation
- ☐ One documented fault: symptom → diagnosis → fix → retest

Student: _____ Date: _____

Instructor sign-off: _____ Score: _____ / 100

Student completion is based on the interactive lab score plus real equipment evidence. Never submit passwords, SNMPv3 secrets, database connection credentials or .env contents.

Official reference sources

LibreNMS installation: https://docs.librenms.org/Installation/Install-LibreNMS/

LibreNMS auto-discovery: https://docs.librenms.org/Extensions/Auto-Discovery/

OPNsense firewall rules: https://docs.opnsense.org/manual/firewall.html

Dell iDRAC manuals: https://www.dell.com/support/manuals/